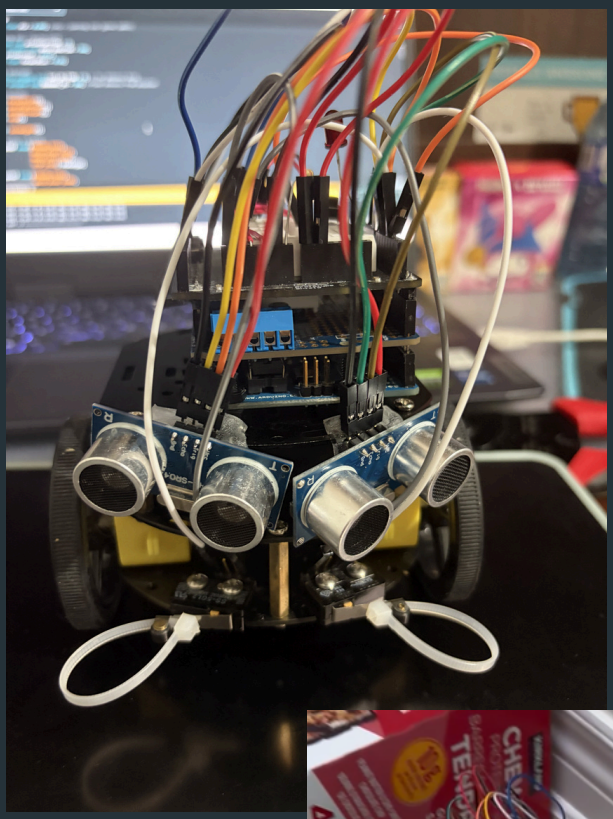
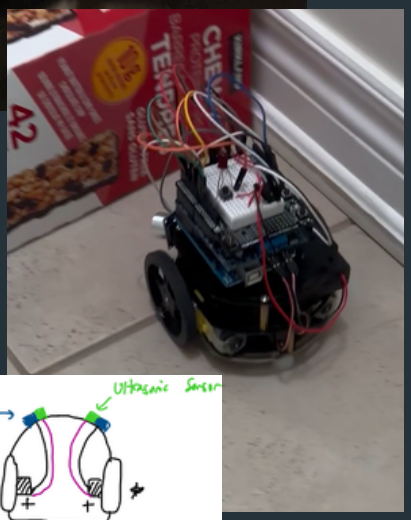
HIBI

AUTONOMOUS MACHINE LEARNING (HEBBIAN LEARNING) OBSTACLE AVOIDANCE ROBOT

Inspired by Chapter 5 of Understanding Intelligence by Rolf Pfeifer, I built Hibi, an autonomous mobile robot designed for obstacle avoidance using proximity and collision sensors. Hibi integrates ultrasonic distance sensors and mechanical collision switches to perceive its environment and control two DC motors through an embedded microcontroller system.

Its behavior is governed by a finite state machine that enables forward motion and avoidance maneuvers, while a neural-network–inspired control structure based on Hebbian learning strengthens associations between proximity sensing and avoidance actions. This allows the robot to gradually rely more on early proximity detection rather than collision events when navigating obstacles.

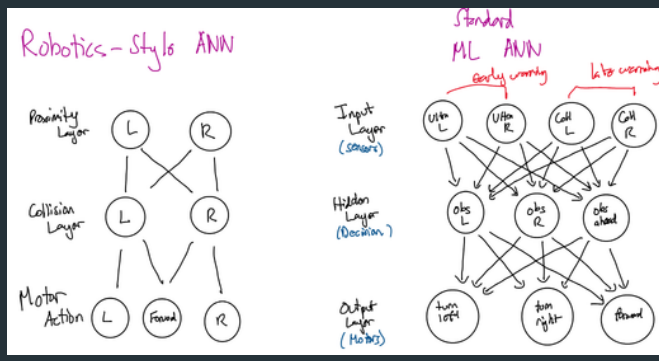


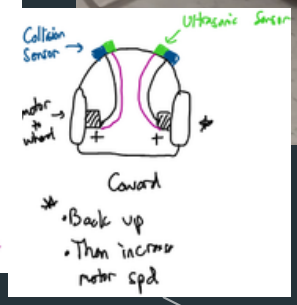


TOOLS TO BUILD HIBI

Limit Switch (SS-5GL2)
Ultrasonic Sonar Distance Sensor (HC-SR04)
Motor Driver Shield (TB6612FNG)
DFRobot Prototype Shield for Uno
DC Gearbox Motors
Arduino UNO
Understanding intelligence Book by Rolf Pfeifer (Chap 5 pg. 139/160)

DIAGRAMS





Watch the Demo Video here: DEMO VIDEO

Or copy: https://drive.google.com/file/d/11S7KAj6Iqm0KiBqjt94TlFJgFTxz4p3S/view?usp=sharing